Example 5.6 The marks obtained by 11 students of a class in a test are 23, 2, 15, 38, 21, 19, 23, 23, 26, 34, 23. Find the mode of the marks.

Solutionv

Arranging the given marks in ascending order, we get,

2, 15, 19, 21, 23, 23, 23, 23, 26, 34, 38.

Clearly, 23 occurs maximum number of times. Hence mode of marks=23.

Example 5.7 Find the mode of the following data 123, 132, 145, 176, 180, 120

Solution

From the above data, we can see that there is no repetition of values in the given data. Each observation occurs only once, so there is no mode.

Note

When each of the observations have occurred only once, then there is no mode for the data.

Think

1. A toy factory making variety of toys for kids, wants to know the most popular toy liked by all the kids. Which average will be the most appropriate for it?
2. Is there a mode exists between the odd numbers from 20 to 40? Discuss.

5.6.1 Mode of large data

When size of the data is large, it is not easy to identify the value which occurs maximum number of times. In that case, we can group the data by using tally marks and then find the mode.

Consider the example to find the mode of the number of goals scored by a football team in 25 matches. The goal scored are 1, 3, 2, 5, 4, 6, 2, 2, 2, 4, 6, 4, 3, 2, 1, 1, 4, 5, 3, 2, 2, 4, 3, 0, 1.

To find the mode of this data, the number of goals score starting from 0 and ending with a maximum of 6 is represented in the form of a table.

Number of goals	Tally marks	Frequency
0	|	1
1	||||	4
2	𝍸 ||	7
3	||||	4
4	𝍸	5
5	||	2
6	||	2
		25

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